

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	KEERTHANA.L
Register Number	192521062

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I KEERTHANA.L-(192521062) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Aim:

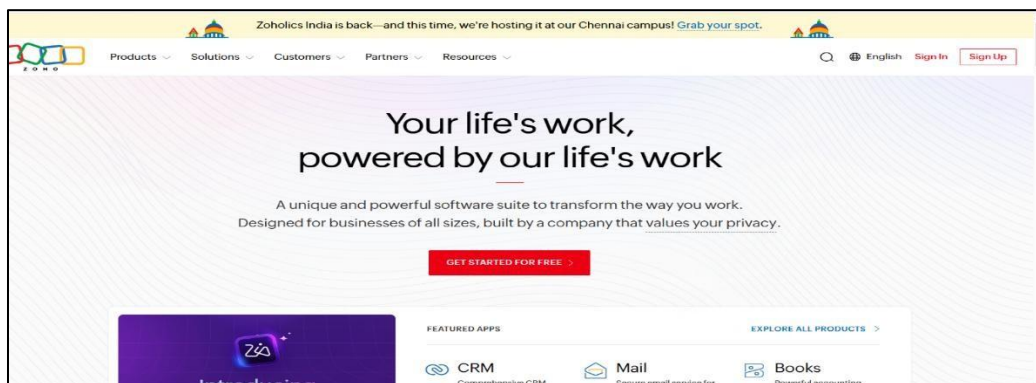
To create a cloud-based Library Book Reservation System using Zoho Creator.

Steps:

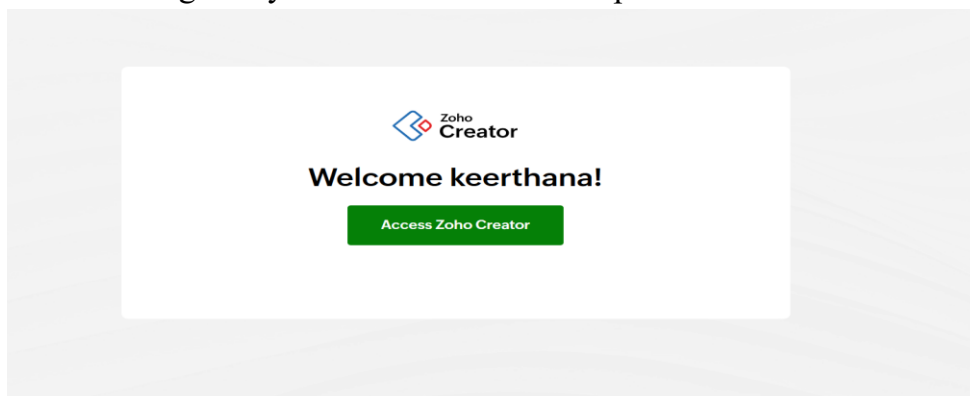
1. Create a new application.
2. Create a Book Reservation form.
3. Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status.
4. Create a report to display available and reserved books.
5. Test the application with sample book records.

Implemented

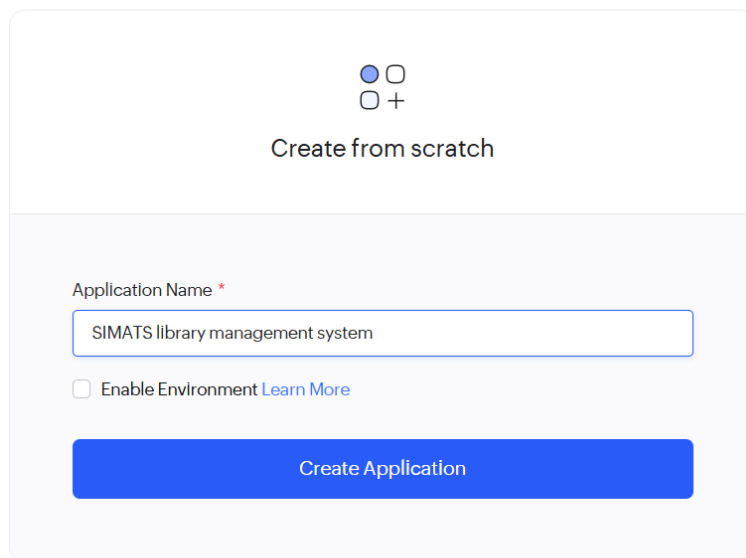
Step 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator

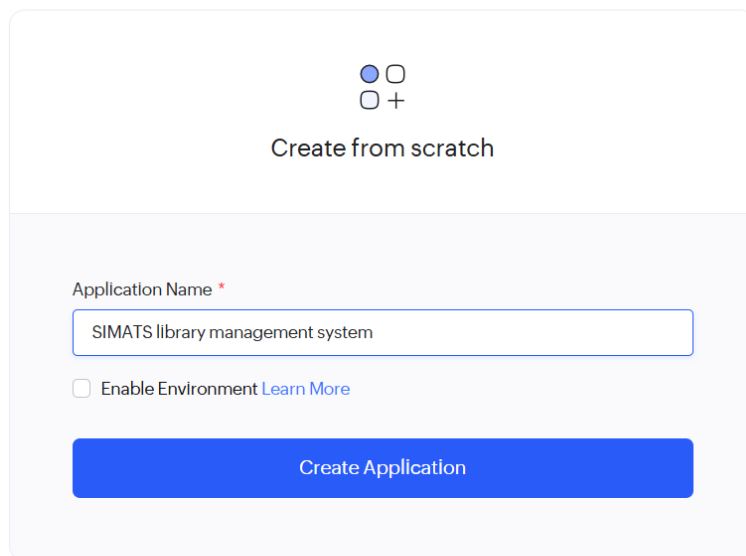


STEP 3: Click Create New Application and select Create from Scratch



The screenshot shows a web interface for creating a new application from scratch. At the top, there is a header with a logo consisting of two circles (one blue, one white) and a plus sign, followed by the text "Create from scratch". Below this, there is a section for "Application Name" with a red asterisk indicating it is required. A text input field contains the text "SIMATS library management system". Below the input field, there is a checkbox labeled "Enable Environment" followed by a blue link "Learn More". At the bottom of the form is a large blue button labeled "Create Application".

STEP 4: Enter Library Book Reservation System as the application name and click Create Application.



This screenshot is identical to the one above, showing the "Create from scratch" form. The "Application Name" field still contains "SIMATS library management system". The checkbox "Enable Environment" and the "Learn More" link are also present. The "Create Application" button is at the bottom.

Step 5: Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status

SIMATS Library Book Reser...

Dashboard

Reservation Details

+ Reservation Details

Reservation Detail...

Board

Book Details

Student Details

keerthana

Reservation Details

Student

Book

Reservation Date

Issue Date

Due Date

Return Date

Fine Amount

Reservation Status

Remarks

dd-MMM-yyyy

dd-MMM-yyyy

dd-MMM-yyyy

dd-MMM-yyyy

#####.##

-Select-

Submit

Reset

SIMATS Library Book Reservation System

SIMATS Library Book Reser...

Dashboard

Reservation Details

Book Details

+ Book Details

Book Details Report

Board

Student Details

keerthana

Book Details

Book Title

Author Name

ISBN Number

Category

Publisher

Edition

Rack Number

Book Status

Book Image

First Name

-Select-

https://

Submit

Reset

SIMATS Library Book Reser...

Dashboard

Reservation Details

Book Details

Student Details

+ Student Details

Student Details Re...

keerthana

Student Details

Student Name

Register Number

Department

Year

Section

Mobile Number

Email Address

Date Registered

Status

First Name

+91 81234 56789

dd-MMM-yyyy

☐ Active

☐ Inactive

☐ Other

STEP 6: Save the form, enter sample booking records, and view them in the Library Book Reservation System.

SIMATS Library Book Reser...

Dashboard

Reservation Details

Reservation Details

Reservation Details Report

Board

Book Details

Student Details

Reservation Details Report

Student	Book	Reservation Date	Issue Date	Due Date	Return Date	Fine Amount	Reservation Status	Remarks
2	1	02-Aug-2026	02-Aug-2026	19-Aug-2026	18-Aug-2026	\$ 0.00	Overdue	Book returne
3	2							
4	3							
5	4							
	5							
1	1	01-Aug-2026	01-Aug-2026	18-Aug-2026	17-Aug-2026	\$ 12.00	Pending	Overdue fine
2	2							
3	3							
1	1	30-Jul-2026	30-Jul-2026	17-Aug-2026	16-Aug-2026	\$ 0.00	Confirmed	Book returne
2	2							
3	3							
1	2	29-Jul-2026	29-Jul-2026	16-Aug-2026	15-Aug-2026	\$ 5.50	Pending	Late return fe
2								
4								
5								
1	1	28-Jul-2026	28-Jul-2026	15-Aug-2026	14-Aug-2026	\$ 0.00	Issued	Book returne
2	2							
3	4							
	5							

Showing 5 of 5

Show Summary

SIMATS Library Book Reservation System

Dashboard

Reservation Details

Book Details

Book Details

Book Details Report

Board

Student Details

Book Details Report

Book ID	Book Title	Author Name	ISBN Number	Category	Publisher	Edition	Rack Number	Book Status	Book
5	Starlight Dreams	Nina Frost	978-1-449-55123-8	Romance	Nova Media	2nd	E-02	Reserved	
4	Urban Legends	Marcus Cole	978-0-306-46397-5	History	City Press	3rd	D-14	Returned	
3	Garden of Secrets	Lily Moon	978-1-506-74812-3	Mystery	Whisper Publishing	1st	C-08	Available	
2	Quantum Horizons	Dr. Aris Thorne	978-0-590-34845-6	Science	Galaxy Books	2nd	B-05	Reserved	
1	The Silent Echo	Elena Vance	978-3-16-148410-0	Fiction	Sunrise Press	1st	A-12	Issued	

SIMATS Library Book Reser...

Dashboard

Reservation Details

Book Details

Student Details

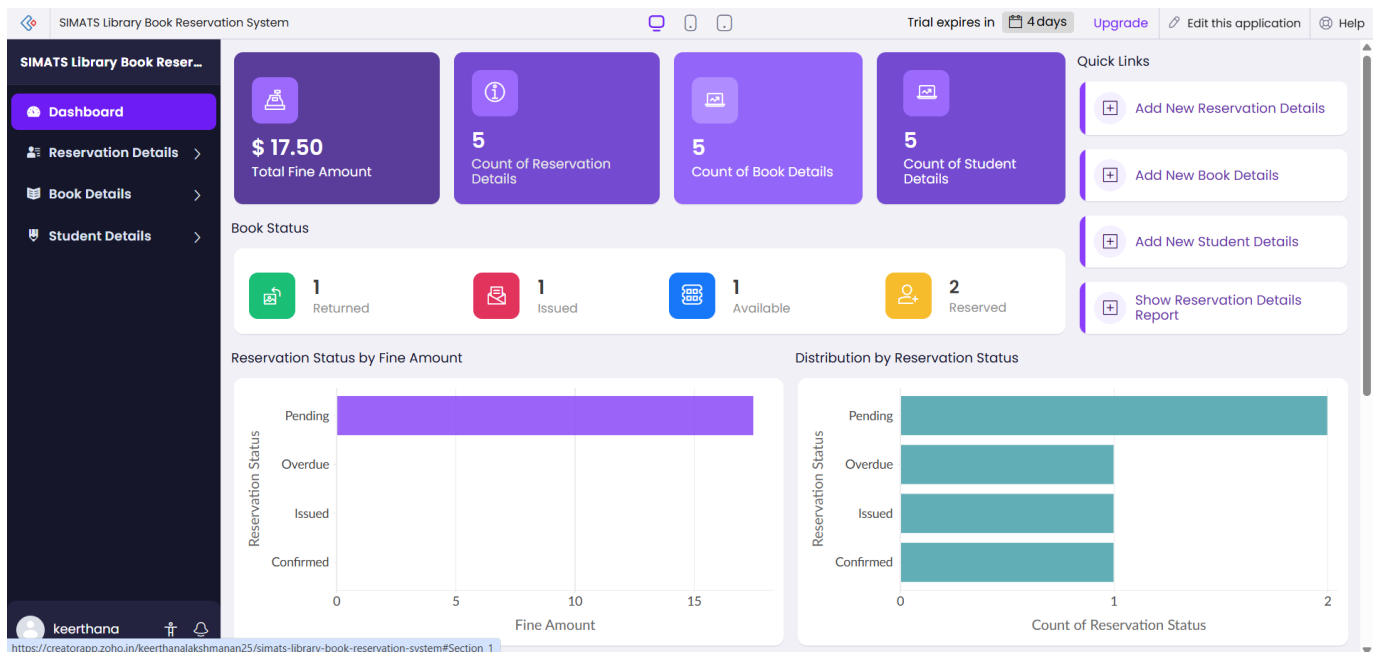
Student Details

Student Details Re...

Student Details Report

Student ID	Student Name	Register Number	Department	Year	Section	Mobile Number	Email Address	Dr
5	Emily Taylor	STU2026005	Industrial Engineering	2026	B	+19738875525	e.taylor@university.edu	26
4	David Martinez	STU2026004	Electrical Engineering	2026	A	+18542479888	d.martinez@university.edu	26
3	Sarah Williams	STU2026003	Civil Engineering	2026	C	+18656432070	s.williams@university.edu	26
2	Michael Chen	STU2026002	Mechanical Engineering	2026	B	+17277356243	m.chen@university.edu	26
1	Alice Johnson	STU2026001	Computer Science	2026	A	+19147482988	alice.j@university.edu	26

Step 7: Final Dashboard



Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

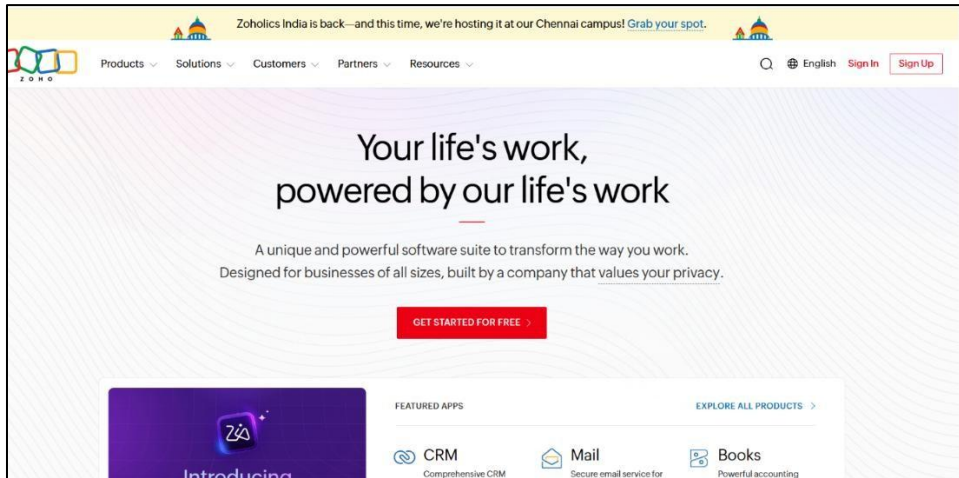
Aim:

To create a cloud-based Payroll Processing System using Zoho Creator.

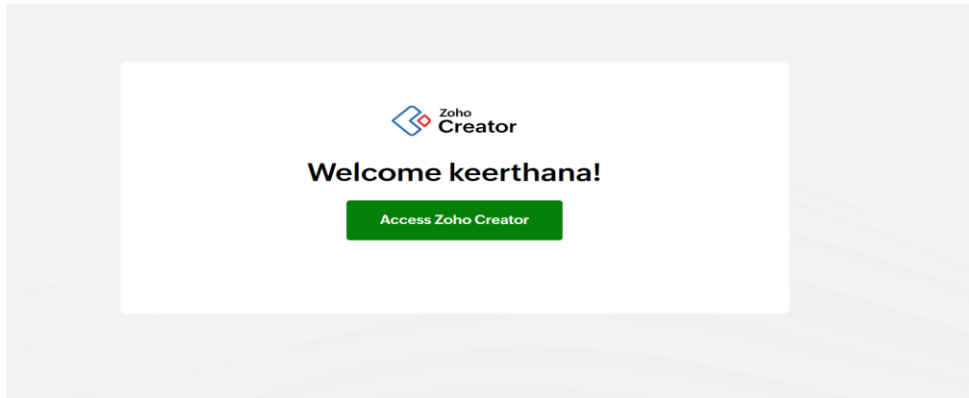
Steps:

1. Create a new application.
2. Create an Employee Payroll form.
3. Add fields: Employee Name, Experience, Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay.
4. Use formulas to calculate Gross Pay and Net Pay.
5. Save employee details and verify the payroll report.

Step 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator



STEP 3: Click Create New Application and select Create from Scratch

The screenshot shows the "Create from scratch" form in Zoho Creator. At the top, there's a header with the text "Create from scratch" and a plus icon. Below this is a form with a label "Application Name *" and a text input field containing "payroll processing system". There's also a checkbox labeled "Enable Environment" with a link "Learn More". At the bottom is a blue button that says "Create Application".

STEP 4: Enter Payroll Processing System as the application name and click Create Application

Create from scratch

Application Name *

payroll processing system

☐ Enable Environment [Learn More](#)

Create Application

Step 5:Add fields: Student Name, Register Number, Subject Marks, Total, Average, Percentage, and Rank.

Payroll Processing System

Dashboard

Employees

+

Employees

Employees Report

Reports

Payrolls

Departments

Dashboards

Department

keerthana

Employees

Employee Name

Department

-Select-

Designation

Mobile Number

+91

81234 56789

Email

Basic Salary

#####.##

Submit

Reset

Payroll Processing System

Dashboard

Employees >

Reports >

+ Reports

Reports Report

Board

Payrolls >

Departments >

Dashboards >

Department >

keerthana

Reports

Report Type

-Select-

Generated Date

dd-MMM-yyyy

Filter Criteria

Report Data

Submit

Reset

Payroll Processing System

Dashboard

Employees >

Reports >

Payrolls >

+ Payrolls

Payrolls Report

Board

Departments >

Dashboards >

Department >

keerthana

Payrolls

Employee

Salary Month

dd-MMM-yyyy

Basic Salary

#####.##

Allowance

#####.##

Deduction

#####.##

Net Salary

#####.##

Payment Status

-Select-

Submit

Reset

Payroll Processing System

Payroll Processing System

Dashboard

Employees >

Reports >

Payrolls >

Departments >

+ Departments

Departments Report

Dashboards >

Department >

keerthana

Departments

Department Name

Manager Name

Submit

Reset

STEP 6: Save the form Payroll Processing, enter sample booking records, and view them in the System.

Payroll Processing System

Trial expires in 4 days Upgrade Edit this application Help

Dashboard

Employees >

Reports >

+ Reports

Reports Report

Board

Payrolls >

Departments >

Dashboards >

Department >

Reports Report

Report Type

Generated Date

Filter Criteria

Report Data

Employee	28-Jul-2026	Overtime Hours	Total overtime logged by department heads
Paid Salary	28-Jul-2026	Leave Balance	Remaining vacation days for each employee
Paid Salary	28-Jul-2026	Bonus Payment	Quarterly bonuses distributed to staff
Paid Salary	28-Jul-2026	Tax Deduction	Tax amounts calculated for all employees
Allowance	28-Jul-2026	Monthly Salary	Total payroll processed for July 2026

Payroll Processing System

Trial expires in 4 days

Upgrade

Edit this application

Help

Payroll Processing System

Dashboard

Employees

Employees

Employees Report

Reports

Payrolls

Departments

Dashboards

Department

Employees Report

Employee ID

Employee Name

Department

Designation

Mobile Number

Email

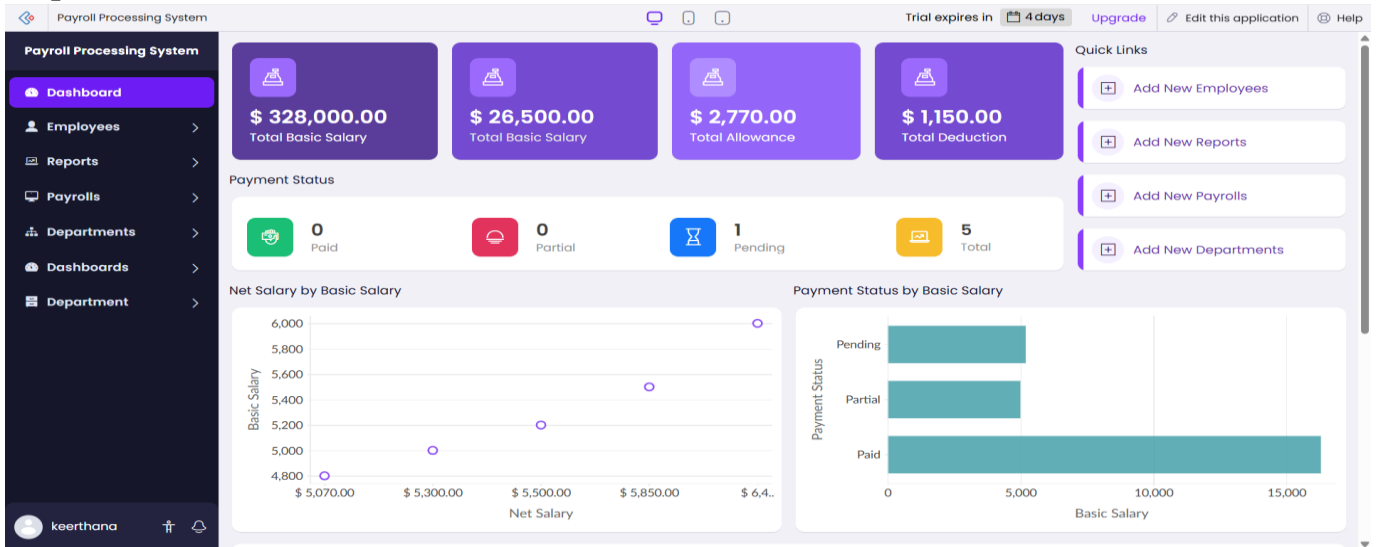
Basic Salary

5	Emma White	Finance and Accounting	Sales Executive	+18326463635	emma.w@company.com	\$ 48,000.00
4	David Lee	Finance and Accounting	Data Analyst	+17034216318	david.l@company.com	\$ 65,000.00
3	Carol Davis	Marketing and Sales	HR Specialist	+19005897908	carol.d@company.com	\$ 58,000.00
2	Bob Williams	Finance and Accounting	Software Engineer	+17608004027	bob.w@company.com	\$ 72,000.00
1	Alice Johnson	Marketing and Sales	Senior Manager	+17259664654	alice.j@company.com	\$ 85,000.00

Showing 5 of 5

Show Summary

Step 7:Final Dashboard



Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

AIM:

To create a virtual machine with 1 cpu, 2gb ram and 15gb storage disk using a type 2 virtualization software.

PROCEDURE:

STEP 1:Download VMware workstation and installed as type 2 hypervisor.

STEP2:Download ubuntu or tiny OS as iso image file.

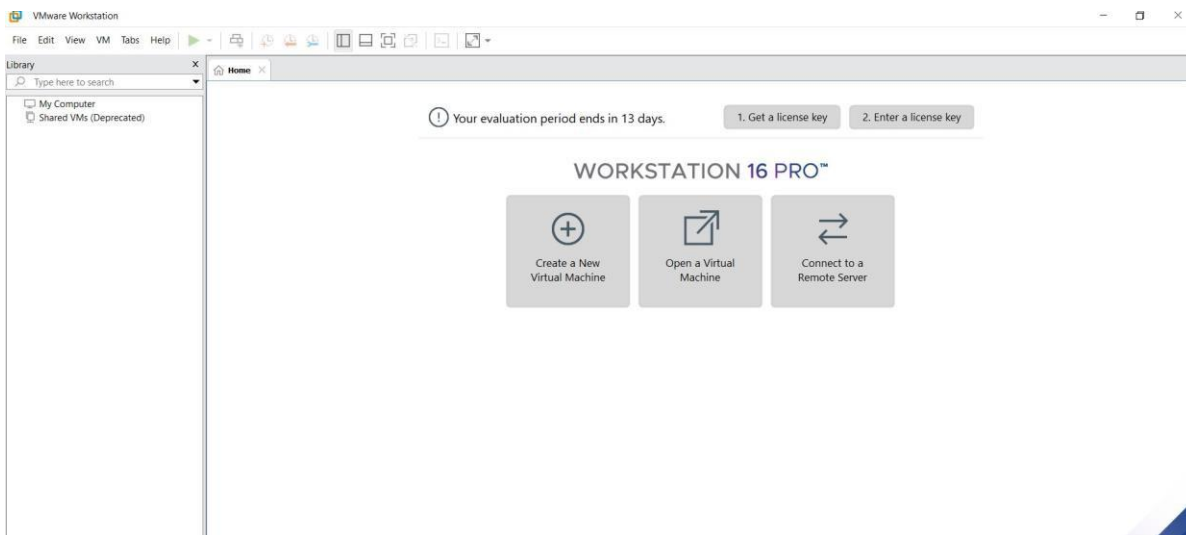
STEP 3: In VMware workstation->create new VM.

STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

STEP 1:DOWNLOAD VMWARE WORKSTATION AND INSTALLED AS TYPE 2 HYPERVISOR

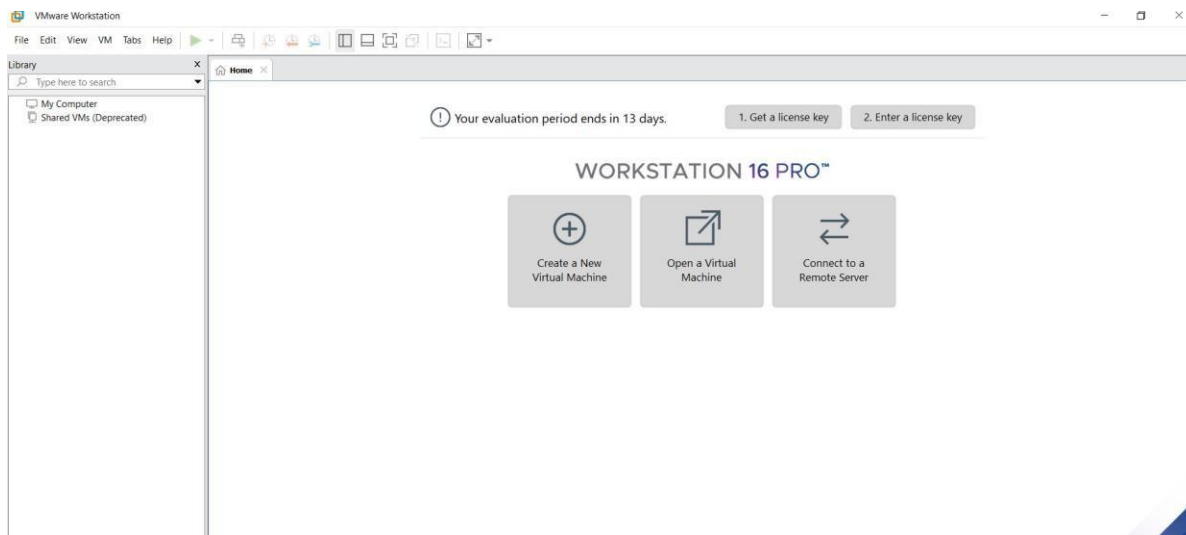


STEP2: DOWNLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

Index of /11.x/x86/release/

../	09-Feb-2020 11:50	-
distribution_files/	03-Dec-2019 11:14	-
src/	01-Apr-2020 07:49	14757888
Core-11.1.iso	01-Apr-2020 07:49	48
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	50639
Core-11.1.iso.zsync	01-Apr-2020 07:49	14757888
Core-current.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso	01-Apr-2020 07:50	52
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	369358
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	216006656
CorePlus-current.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso	01-Apr-2020 07:50	52
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	68301
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	19922944
TinyCore-current.iso		

STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM



STEP 4: DO THE BASIC CONFIGURATION SETTINGS.



! Your evaluation period ends in 12 days.

1. Get a license key

2. Enter a license key

New Virtual Machine Wizard x 6 PRO™

Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

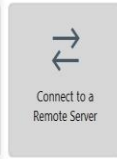
Install from:

☐ Installer disc:
No drives available

☐ Installer disc image file (iso):
C:\Users\Gopi Prasad-PC\Downloads\ubuntu-20.04.2... Browse...

☒ I will install the operating system later.
The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel



New Virtual Machine Wizard x 6 PRO™

Select a Guest Operating System
Which operating system will be installed on this virtual machine?

Guest operating system

☐ Microsoft Windows

☒ Linux

☐ VMware ESX

☐ Other

Version

Ubuntu 64-bit

Help < Back Next > Cancel



New Virtual Machine Wizard x 6 PRO™

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:
Ubuntu Vmware

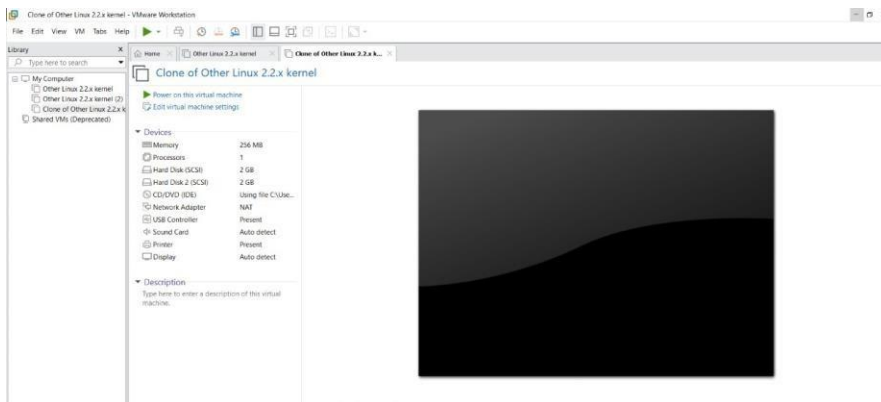
Location:
C:\Users\Gopi Prasad-PC\Documents\Virtual Machines\Ubuntu... Browse...

The default location can be changed at Edit > Preferences.

< Back Next > Cancel



STEP 5: CREATED TINYOS VIRTUAL MACHINE



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux).

PROCEDURE

Download VMware workstation and installed as type 2 hypervisor.

STEP2:Download ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

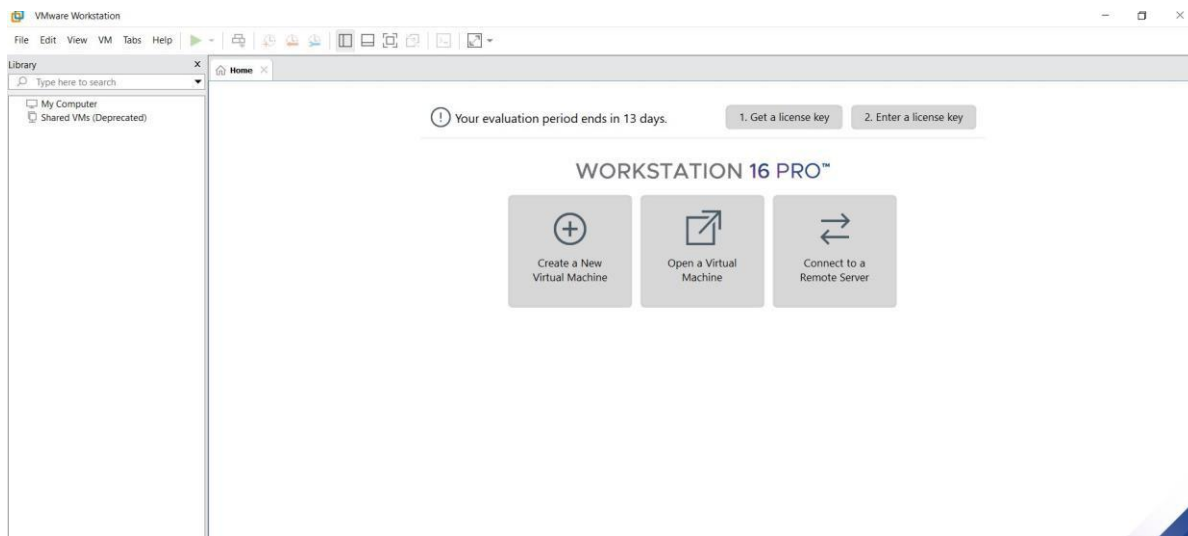
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

IMPLEMENTATION:

STEP 1:DOWNLOAD VMWARE WORKSTATION AND INSTALLED AS TYPE 2 HYPERVISOR

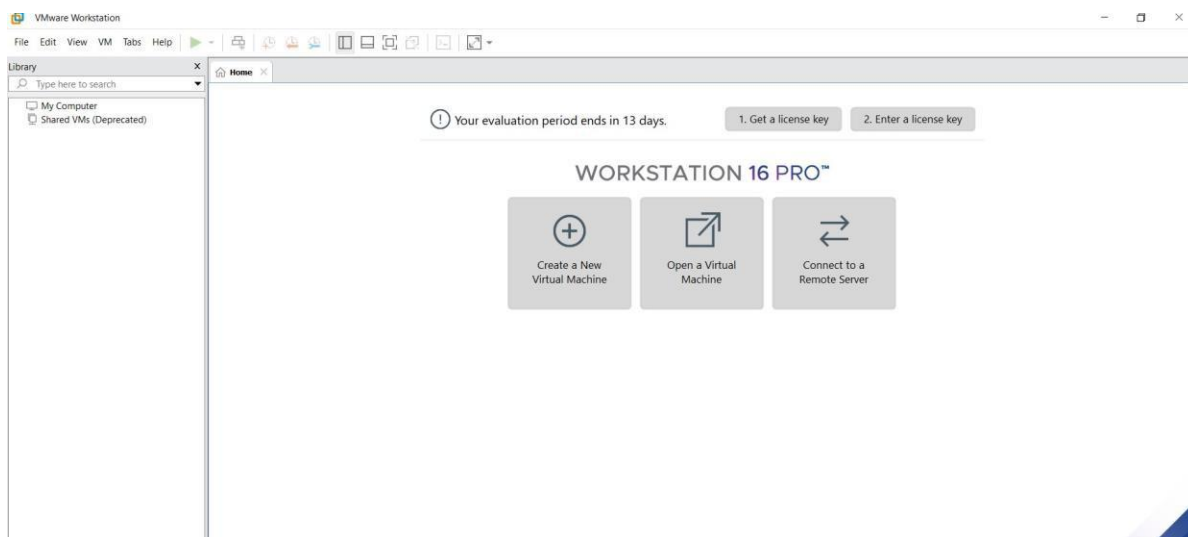


STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

Index of /11.x/x86/release/

distribution_files/	09-Feb-2020 11:50	-
src/	03-Dec-2019 11:14	-
Core-11.1.iso	01-Apr-2020 07:49	14757888
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	48
Core-11.1.iso.xsync	01-Apr-2020 07:49	50639
Core-current.iso	01-Apr-2020 07:49	14757888
CorePlus-11.1.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	52
CorePlus-11.1.iso.xsync	01-Apr-2020 07:50	369358
CorePlus-current.iso	01-Apr-2020 07:50	216006656
TinyCore-11.1.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	52
TinyCore-11.1.iso.xsync	01-Apr-2020 07:50	68301
TinyCore-current.iso	01-Apr-2020 07:50	19922944

STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM



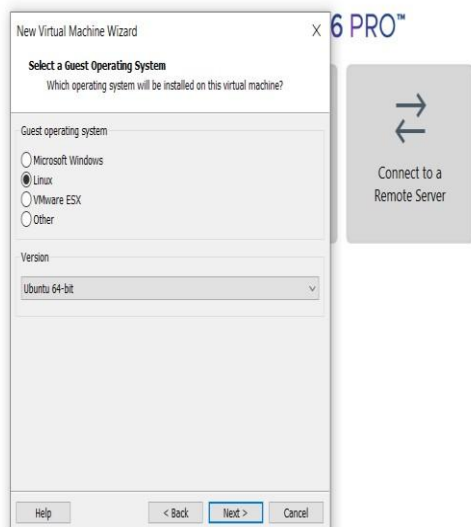
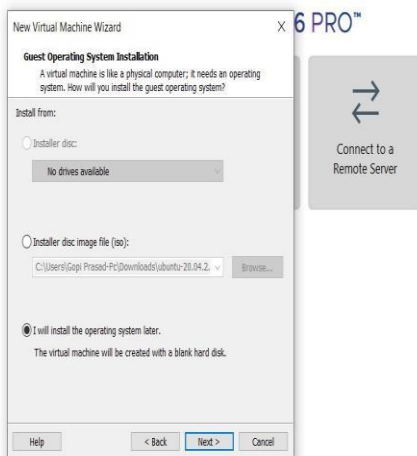
STEP 4: DO THE BASIC CONFIGURATION SETTINGS

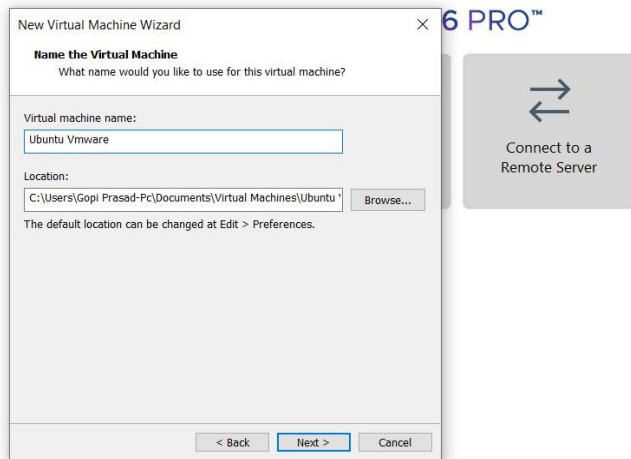


! Your evaluation period ends in 12 days.

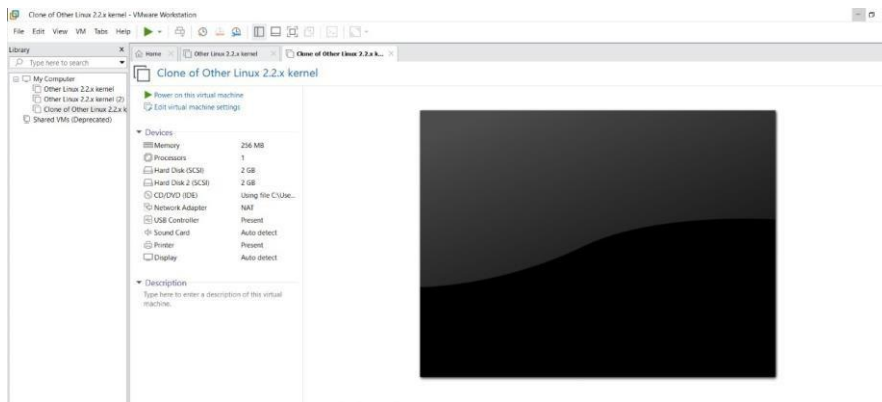
1. Get a license key

2. Enter a license key





STEP 5: CREATED TINYOS VIRTUAL MACHINE



STEP 6: LAUNCH THE VM

